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Wind load Analysis For Rolling Shutter

as per Indian Standards IS 875 Part III 1987

Project: ALSTOM
Client: TATA PROJECT
Date: 03/06/2017
Height: 6000 mm
Width: 4000 mm
Slat Size: 95 mm
Slat Width: 19 mm

Wind Load Analysis as per IS

Design Wind Speed Vz= Vb * k1 * k2 * k3
where Vb = Basic Wind Speed in m / s.
k1 = Risk Coefficient (clause 5.) = 1
k2 = THS Factor (clause 5.3 = 1
k3 = Topography Factor (clause 5.) = 1
Vb = Maximum = 47 m/s

Wind Speed Vz = 47 m/s

Terrain Category: 2
(Open terrain with well-scattered obstructions having heights between 1.5m to 10m)

Class of Building: A(Structures having max. dimensions is less then 20m)

k1, k2 & k3 constants are derived from IS-875, based on location height of structure & geographical factor. These factors are required for tall structures, Where we need to consider varying pressure condition.

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Wind Load Analysis as per IS 875 - III

Design wind pressure = $0.6 \times V_z^2$

Where,

Pz = Design wind pressure

Vz = Design wind velocity

Wind Pressure, Pz = 0.6 x (47)2

Wind Pressure, Pz = 1325.4 N/m^2

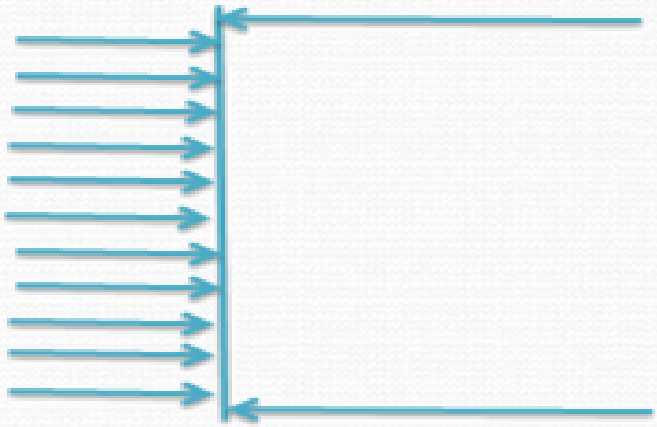
Pz = 0.001325 N/mm^2

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Wind Load Analysis as per IS 875 - III

Case I - Simply supported beam with uniform pressure.

1325 N/m^2



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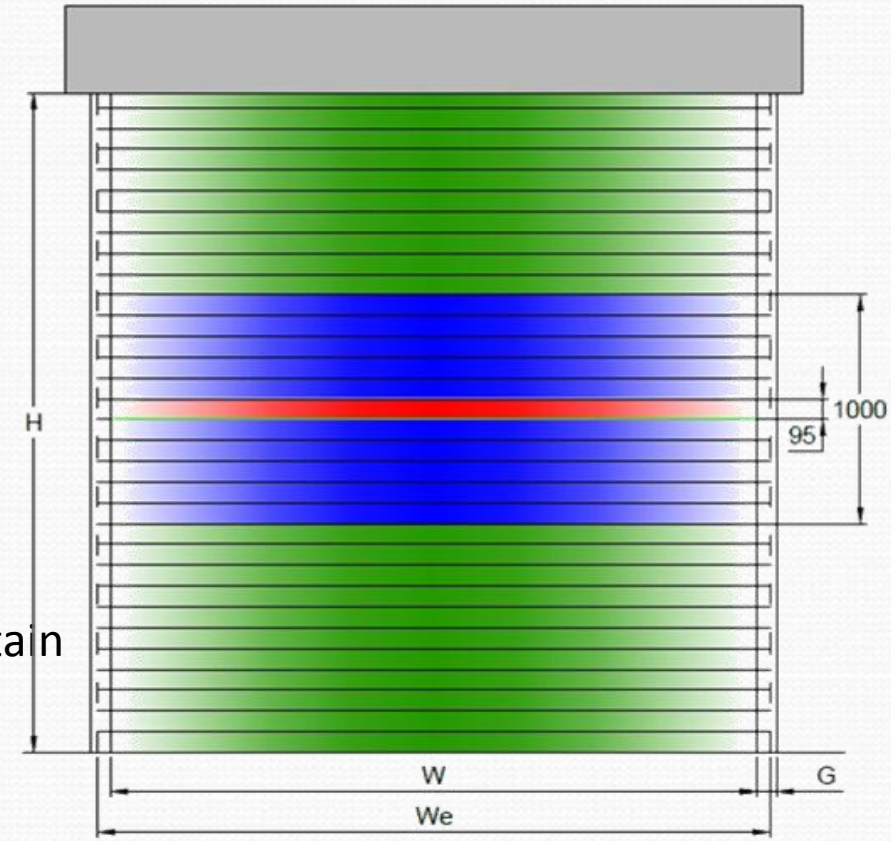
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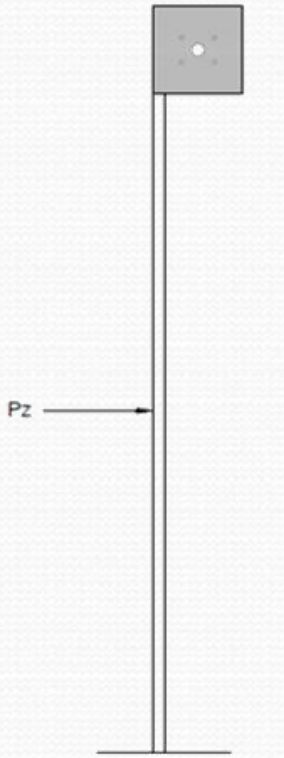
A)Bending Moment

B)Deflection

C)Maximum Stress

D)Side guides for curtain





Pz- Design pressure
W- Width of shutter
We- Effective width of shutter
H- Height of shutter
G- Width of guide

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A)Bending Moment

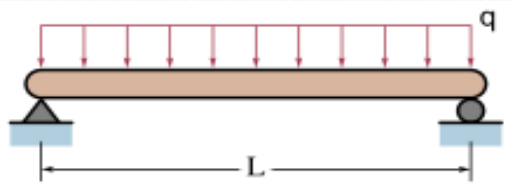
Door is simply supported on two opposite sides

Max B.M. = Design Pressure x $We^2/8$

B)Deflection

Max. deflection will be at the centre

D max = (5x M x We^2) / (48 x E x I)



Where ,

M = Bending Moment of 75mm Curved type

E = Modules of Elasticity for CRS = 210GPa

I = Moment of Inertia of slat (b x d3 /12)

We= Effective Width of Shutter

w = Clear Width of shutter

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C)Max Stress

Max. Stress= W_e^2xpressure on stripx1000xC/12x I

Where,

C= centre of gravity ,I= moment of inertia ,We= effective width

Permissible Stress =248/FOS

=248/1.5

= 165 N/sq.mm

D)Side Guides for Curtain panel

Side Guides required = w/60

Where,

w= width of shutter

Design calculations for shutter of
Size: 4000mm(w) x 6000mm(H)

1. Wind Analysis for rolling shutter of size 4000mm(w) x 6000mm(h)

ØDesign Wind Speed Vz= Vb * k1 * k2 * k3

where,

Vb = Basic Wind Speed in m / s.

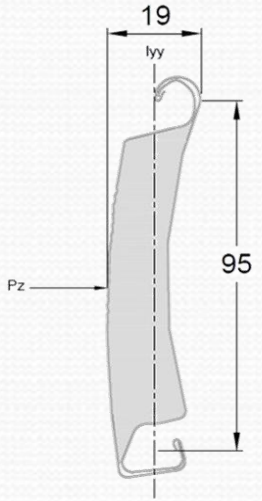
k1 = Risk Coefficient (clause 5.4.1) = 1

k2 = THS Factor (clause 5.3.2) = 1

k3 = Topography Factor (clause 5.3.3) = 1

Vb = Maximum = 47

wind Speed Vz = 47 m/s



(of 1 meter slat)

Design wind pressure = 0.6x 47x47= 1325.4 N/m2

Bending Moment M_1 = ((Design pressure x Width^2)/8

= 0.0013254 x 4000 x 4000 / 8

= 2650.80 N-mm

Bending Moment M = M_1x0.095m (Moment of 95mm slat)

= 2650.8 x 95

= 251.83 N-mm

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Ø Assuming thickness of slat 0.5 mm & minimum height as 95mm & depth as 19mm.

(From CAD DATA) I= 8450 mm^4

E= modulus of elasticity for CRS= 210 GPA

= 210000 N/mm^2

Max. deflection = 5 x 2650.8 x 4000 x 4000 / (48 x 210000 x 8450)

= 2.49 mm

Permissible deflection = 4000/250

As per IS:800 = 16 mm

Deflection is within limit.

Max. stress = = 251.83 x (19 / 2)/ 8450

= 0.28 N/mm^2

Permissible stress = (248 N/mm2) /1.5

= 165 N/mm^2

Hence, design is Safe on conditions.

Guide side required = 4000/60 = 66.67

Hence we are using 100mm guide.

Thank You

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